et al. (2019) studies gradients propagation through the depth. Indeed, our analysis implies that including BN imposes the Gaussian representations required for these analyses. Although our result is established for linear activations, we conjecture a similar result holds for non-linear MLPs (see Appendix F).

5. The orthogonality and optimization

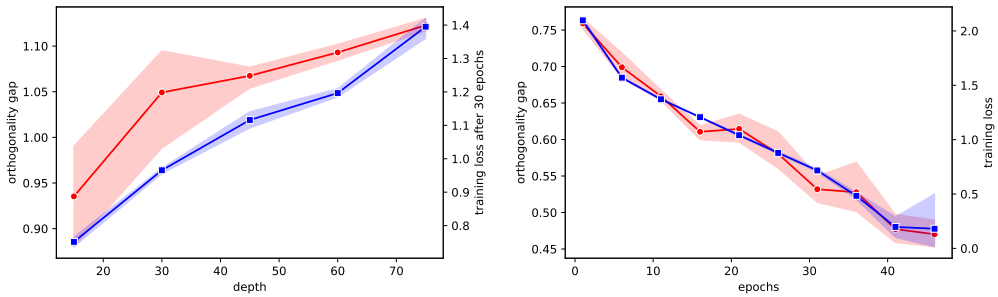
In the preceding sections, we elaborated on the theoretical properties of BN networks in controlled settings. In this section, we demonstrate the practical applications of our findings. In the first part, we focus on the relationship between depth and orthogonality. Increasing depth drastically slows the training of neural networks with BN. Furthermore, we observe that as depth grows, the training slowdown highly correlates with the orthogonality gap. This observation suggests that SGD needs to orthogonalize deep representations in order to start classification. This intuition leads us to the following question: If orthogonalization is a prerequisite for training, can we save optimization time by starting from orthogonal representations? To test this experimentally, we devised a weight initialization that guarantees orthogonality of representations. Surprisingly, even in a network without BN, our experiments showed that this initialization avoids the training slow down, affirmatively answering the question.

Throughout the experiments, we use vanilla MLP (without BN) with a width of 800 across all hidden layers, ReLU activation, and used Xavier's method for weights intialization (Glorot and Bengio, 2010). We use SGD with stepsize 0.01 and batch size 500 and for training. The learning task is classification with cross entropy loss for CIFAR10 dataset (Krizhevsky et al., 2009, MIT license). We use PyTorch (Paszke et al., 2019, BSD license) and Google Colaboratory platform with a single Tesla-P100 GPU with 16GB memory in all the experiments. The reported orthogonality gap is the average of the orthogonality gap of representation in the last layer.

5.1. **Orthogonality correlates with optimization performance.** In the first experiment, we show that the orthogonality of representations at the initialization correlates with optimization speed. For networks with 15, 30, 45, 60, and 75 widths, we register training loss after 30 epochs and compare it with the initial orthogonality gap. Figure 3a shows the training loss (blue) and the initial orthogonality gap (red) as a function of depth. We observe that representations are more entangled, i.e., orthogonal, when we increase depth, coinciding with the training slowdown. Intuitively, the slowdown is due to the additional time SGD must spend to orthogonalize the representations before classification. In the second experiment, we validate this intuitive argument by tracking the orthogonality gap during training. Figure 3b plots the orthogonality gap of output and training loss for a network with 20 layers. We observe that SGD updates are iteratively orthogonalizing representations, marked by the reduction in the orthogonality gap.

5.2. **Learning with initial orthogonal representations.** We have seen that the slowdown in SGD for deeper networks correlates with the orthogonality gap before training. Here we show that by preemptively orthogonalizing representations, we avoid the slowdown with depth. While in MPL with linear activations, hidden representations remain orthogonal simply by taking orthogonal matrices as weights (Pennington et al., 2018; Saxe et al., 2014), the same does not hold for networks with non-linear activations, such as ReLU. To enforce the orthogonality in the absence of BN, we introduce a dependency between weights of successive layers that ensures deep representations remain orthogonal. More specifically, we incorporate the SVD decomposition of the hidden representation of each layer into the initialization of the subsequent layer. To emphasis this dependency between layers and to distinguish it from purely orthogonal weight initialization, we refer to this as *iterative orthogonalization*.

We take a large batch of samples $n \geq d$, as the input batch for initialization. Let us assume that weights are initialized up to layer $W_{\ell-1}$. To initialize W_ℓ, we compute SVD decomposition of the



(A) gap and loss vs. depth (B) gap and loss during training

FIGURE 3. **Orthogonality and Optimization** Left: the orthogonality gap at initialization (red, left axis) and the training loss after 30 epochs (blue, right axis) with depth. Right: the orthogonality gap (red, left axis) and the training loss in each epoch (blue, right axis). Mean and 95% confidence interval of 4 independent runs.

representations $H_\ell = U_\ell \Sigma_\ell V_\ell^\top$ where matrices $U_\ell \in \mathbb{R}^{d\times d}$ and $V_\ell \in \mathbb{R}^{n\times d}$ are orthogonal. Given this decomposition, we initialize W_ℓ by

$$W_\ell = \frac{1}{\|\Sigma_\ell^{1/2}\|_F} V_\ell' \Sigma_\ell^{-1/2} U_\ell^\top, \tag{6}$$

where $V_\ell' \in \mathbb{R}^{d\times d}$ is an orthogonal matrix obtained by slicing $V_\ell \in \mathbb{R}^{n\times d}$. Notably, the inverse in the above formula exists when n is sufficiently larger than d [2]. It is easy to check that $V(W_\ell H_\ell) < V(H_\ell)$ holds for the above initialization (see Appendix E), similar to BN. By enforcing the orthogonality, this initialization significantly alleviates the slow down of training with depth (see Fig. 4), with no need for BN. This initialization is not limited to MLPs. In Appendix G, we propose a similar SVD-based initialization for convolutional networks that effective accelerates training of deep convolutional networks.

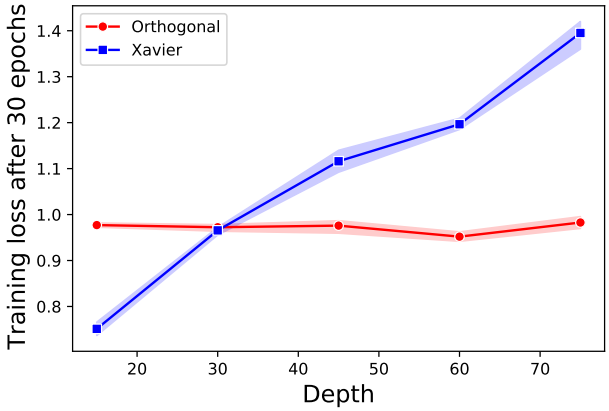


FIGURE 4. **Iterative orthogonalization.** Horizontal axis: depth. Vertical axis: the training loss after 30 epochs for Xavier's initialization (blue), our initialization (red). Mean and 95% confidence interval of 4 independent runs.

[2]We may inductively assume that H_ℓ is almost orthogonal by the choice of $W_1, \ldots, W_{\ell-1}$. Thus, Σ_ℓ is invertible.

6. Discussion

To recap, we proved the recurrence of random linear transformations and BN orthogonalizes samples. Our experiments underline practical applications of this theoretical finding: starting from orthogonal representations effectively avoids the training slowdown with depth for MLPs. Based on our experimental observations in Appendix G, we believe that this result extends to standard convolution networks used in practice. In other words, a proper initialization ensuring the orthogonality of hidden representations may replace BN in neural architectures. This future research direction has the potentials to boost the training of deep neural networks and change benchmarks in deep learning.

Although our theoretical bounds hold for MLPs with linear activations, our experiments confirm similar orthogonal stability for various neural architectures. Appendix Figure 5 demonstrates this stability for MLPs with ReLU activations and a convolutional networks. This plot compares the evolution of the orthogonality gap, through layers, for BN networks with vanilla networks. For vanilla networks, we observe that the gap increases with depth. On the contrary, the gap decreases by adding BN layers and stabilizes at a term that is constant with regard to depth. Based on our observations, we conjecture that hidden representations of modern BN networks obey similar stability. A more formal statement of the conjecture is presented in Appendix F.

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